CS601 – Machine Learning

Unit – I

- **Main**
  - Introduction to Machine Learning
  - Mathematical Foundations
  - Learning Models
- **Detail**
  - Scope and limitations
  - Regression
  - Probability – statistics – linear algebra for machine learning
  - Convex optimization
  - Data visualization
  - Hypothesis function and testing
  - Data distributions
  - Data preprocessing – data augmentation – normalizing data sets
  - Machine learning models
  - Supervised – unsupervised learning

Unit – II

- **Main**
  - Neural Network Basics
  - Training Mechanisms
  - Regularization
- **Detail**
  - Linearity vs non linearity
  - Activation functions (sigmoid – ReLU – etc.)
  - Weights and bias
  - Loss function
  - Gradient descent
  - Multilayer network – backpropagation
  - Weight initialization
  - Training – testing
  - Unstable gradient problem
  - Auto encoders
  - Batch normalization – dropout
  - L1 – L2 regularization
  - Momentum – tuning hyper parameters

Unit – III

- **Main**
  - Convolutional Neural Network
  - CNN Layers

    - CNN Implementation
- **Detail**
    - Convolutional neural network
    - Flattening – subsampling – padding – stride
    - Convolution layer – pooling layer – loss layer – dense layer
    - 1×1 convolution
    - Inception network
    - Input channels
    - Transfer learning – one shot learning
    - Dimension reductions
    - CNN implementation (TensorFlow – Keras)

Unit – IV

- **Main**
    - Recurrent Neural Network
    - Sequence Models
    - Reinforcement Learning
- **Detail**
    - Recurrent neural network
    - Long short-term memory
    - Gated recurrent unit
    - Translation
    - Beam search and width
    - BLEU score
    - Attention model
    - Reinforcement learning framework
    - MDP – Bellman equations
    - Value iteration – policy iteration
    - Actor-critic model
    - Q-learning – SARSA

Unit – V

- **Main**
    - Classical ML Algorithms
    - Bayesian Learning
    - Applications
- **Detail**
    - Support vector machines
    - Bayesian learning
    - Applications in computer vision
    - Speech processing
    - Natural language processing
    - Case study – ImageNet competition

CS602 – Computer Networks

Unit – I

- **Main**
  - Computer Network Basics
  - Layered Architecture
  - OSI Model
- **Detail**
  - Network definitions – goals – components – architecture – classifications
  - Protocol hierarchy – interfaces – services
  - Connection oriented – connectionless services
  - Service primitives
  - ISO-OSI reference model
  - Comparison with TCP/IP
  - Physical layer principles (media – bandwidth – data rate – modulation)

Unit – II

- **Main**
  - Data Link Layer
  - Error and Flow Control
  - ARQ Protocols
- **Detail**
  - DLL need – services – framing
  - Flow control – error control
  - Sliding window protocols
  - Go-Back-N – Selective Repeat – Hybrid ARQ
  - Protocol verification
  - Finite state machine – Petri net
  - ARP – RARP – GARP

Unit – III

- **Main**
  - MAC Sublayer
  - Access Protocols
  - IEEE Standards
- **Detail**
  - MAC addressing
  - Binary exponential back-off algorithm
  - ALOHA – Slotted ALOHA
  - CSMA – CSMA/CD – CSMA/CA
  - Collision free protocols
  - Performance measuring metrics
  - IEEE 802 series standards

Unit – IV

- **Main**
  - Network Layer
  - Routing Algorithms
  - IP Protocol
- **Detail**
  - Network layer services – design issues
  - Dijkstra – Bellman-Ford
  - Hierarchical – broadcast – multicast routing
  - IP addressing – header format
  - Packet forwarding
  - Fragmentation – reassembly
  - ICMP
  - IPv4 vs IPv6

Unit – V

- **Main**
  - Transport Layer
  - Application Layer
  - Network Management
- **Detail**
  - UDP (header – checksum – multicast traffic)
  - TCP (connection management – reliability – flow control – congestion control – timers)
  - Application layer protocols (HTTP – FTP – SSH – SMTP – MIME – IMAP – DNS)
  - SNMP

CS603 (A) – Advanced Computer Architecture

Unit – I

- Main
  - Parallel Computer Models
  - Performance Factors
  - Interconnection Networks
- Detail
  - Flynn's classification
  - Hardware – software parallelism
  - Program partitioning – scheduling
  - Static – dynamic interconnection networks

Unit – II

- Main
  - Instruction Set Architecture
  - Memory Hierarchy
  - Bus Systems
- Detail
  - CISC – RISC – VLIW
  - Memory coherence – locality
  - Interleaved memory
  - Backplane bus systems

Unit – III

- Main
  - Pipeline Processors
  - Superscalar Design
  - Scheduling
- Detail
  - Linear – nonlinear pipelines
  - Pipeline hazards
  - Scoreboarding – Tomasulo algorithm
  - Branch handling
  - Arithmetic pipelines

Unit – IV

- Main
  - Cache Coherence

    - Vector Processing
    - Multithreading
- **Detail**
    - Snoopy – directory protocols
    - SIMD organization
    - Vector supercomputers
    - Multithreading issues

Unit – V

- **Main**
    - Parallel Programming Models
    - Languages
    - Tools
- **Detail**
    - Shared variable – message passing – data parallel
    - Functional – logic models
    - Parallel compilers
    - Programming environments

CS603 (B) – Computer Graphics & Visualization

Unit – I

- **Main**
  - Raster Graphics
  - Drawing Algorithms
  - Polygon Filling
- **Detail**
  - Raster scan displays – frame buffer
  - DDA – Bresenham line algorithms
  - Circle drawing algorithms
  - Boundary fill – flood fill

Unit – II

- **Main**
  - 2D Transformations
  - Windowing
  - Clipping
- **Detail**
  - Translation – rotation – scaling – shearing
  - Homogeneous coordinates
  - Line – polygon clipping

Unit – III

- **Main**
  - 3D Transformations
  - Projection
  - Illumination
- **Detail**
  - Parallel – perspective projection
  - Hidden surface elimination
  - Z-buffer – painter's algorithm
  - Shading models

Unit – IV

- **Main**
  - Visualization Techniques
  - Volume Rendering
  - Animation

- **Detail**
  - Scalar – vector field visualization
  - Ray casting
  - Time varying data
  - Animation techniques

Unit – V

- **Main**
  - Multimedia
  - Animation
  - Compression
- **Detail**
  - Text – audio – video
  - Multimedia formats
  - Lossless – lossy compression
  - MPEG standards

CS603 (C) – Compiler Design

Unit – I

- **Main**
  - Compiler Structure
  - Lexical Analysis
  - LEX
- **Detail**
  - Compiler phases
  - Token recognition
  - Lexical analyzer generator

Unit – II

- **Main**
  - Syntax Analysis
  - Parsing Techniques
  - Syntax Directed Translation
- **Detail**
  - Top-down – bottom-up parsing
  - LR parsers
  - Syntax trees – attributes

Unit – III

- **Main**
  - Type Checking
  - Runtime Environment
  - Symbol Table
- **Detail**
  - Type systems
  - Storage allocation
  - Error detection

Unit – IV

- **Main**
  - Intermediate Code
  - Code Generation
  - Optimization
- **Detail**
  - DAG representation
  - Register allocation
  - Peephole optimization

Unit – V

- **Main**
  - Code Optimization
  - Data Flow Analysis
  - Debugging
- **Detail**
  - Loop optimization
  - Dead code elimination
  - Global optimization

CS604 (A) – Knowledge Management (Open Elective)

Unit – I

- **Main**
  - Knowledge Management
  - Foundations
  - Ethics
- **Detail**
  - Introduction to knowledge management
  - Foundations (cultural issues – technology applications – organizational concepts – processes – management aspects – decision support systems)
  - Evolution of knowledge management
  - Information management to knowledge management
  - Key challenges
  - Ethics for knowledge management

Unit – II

- **Main**
  - Organization
  - Learning Culture
  - Knowledge Sharing
- **Detail**
  - Organization and knowledge management
  - Building the learning organization
  - Knowledge markets
  - Cooperation among distributed technical specialists
  - Tacit knowledge
  - Quality assurance

Unit – III

- **Main**
  - KM Tools
  - Information Technology
  - Vocabulary Control
- **Detail**
  - Telecommunications and networks in KM
  - Internet search engines and KM
  - IT support for KM
  - Knowledge management and vocabulary control
  - Information mapping
  - Information coding

    - Repackaging information

Unit – IV

- **Main**
    - Knowledge Strategy
    - KM Applications
    - Case Studies
- **Detail**
    - Components of a knowledge strategy
    - From library to knowledge center
    - Knowledge management in health sciences
    - Knowledge management in developing countries

Unit – V

- **Main**
    - Future Trends
    - KM Planning
    - Corporate Memory
- **Detail**
    - Advanced topics in KM
    - Knowledge management map/plan
    - Integration with organizational strategy
    - Corporate memories case study

CS604 (B) – Rural Technology & Community Development (Open Elective)

Unit – I

- Main
  - Rural Management
  - Management Principles
  - Decision Making
- Detail
  - Introduction to management
  - Theory of management
  - Planning
  - Organisation structure and design
  - Motivation and leadership
  - Management control
  - Managerial decision making

Unit – II

- Main
  - Human Resource Management
  - Rural Workforce
  - Industrial Relations
- Detail
  - Nature and scope of HRM
  - Human resource planning
  - Recruitment and selection
  - Training and development
  - Performance appraisal
  - Welfare programmes
  - Fringe benefits
  - Wage and salary administration
  - Morale and productivity
  - Industrial relations
  - Industrial disputes

Unit – III

- Main
  - Rural Financing
  - Agricultural Credit
  - Government Institutions
- Detail
  - Rural credit system

    - Role of rural credit in rural development
    - Evolution of rural credit system in India
    - Agricultural credit
    - Agricultural credit review committee
    - Reports of committees and commissions
    - Problems and prospects
    - Rural credit to non-farm sector
    - Credit for small and marginal entrepreneurs
    - Role of government institutions
    - Role of non-government / semi-government / quasi-government institutions
    - Financing for small scale and cottage industries

Unit – IV

- **Main**
    - Research Methodology
    - Data Collection
    - Qualitative Methods
- **Detail**
    - Concept of social research
    - Traditional research
    - Action research
    - Participatory research
    - Qualitative data construction
    - Methods of data collection
    - Interview techniques
    - Sociometry
    - Case studies
    - Observation
    - Coding
    - Content analysis

Unit – V

- **Main**
    - Data Analysis
    - Statistical Measures
    - Sampling
- **Detail**
    - Collection of data
    - Tabulation
    - Presentation of data
    - Measures of central tendency
    - Dispersion
    - Moments
    - Skewness

- Kurtosis
- Correlation
- Regression
- Sampling theory
- Test of significance

CS604 (C) – Computer Graphics / Multimedia (as per syllabus option)

Unit – I

- Main
  - Multimedia Basics
  - Applications
  - Text
- Detail
  - Basics of multimedia
  - Applications of multimedia
  - Text types
  - Unicode standard
  - Text compression
  - Text file formats

Unit – II

- Main
  - Audio
  - Digital Audio
  - Audio Processing
- Detail
  - Audio components
  - Digital audio
  - Digital audio processing
  - Sound cards
  - Audio file formats
  - Audio processing software

Unit – III

- Main
  - Video
  - Digital Video
  - Video Processing
- Detail
  - Video color spaces
  - Digital video
  - Digital video processing
  - Video file formats

Unit – IV

- **Main**
  - Animation
  - 3D Animation
  - Special Effects
- **Detail**
  - Uses of animation
  - Principles of animation
  - Computer based animation
  - 3D animation
  - Animation file formats
  - Animation software
  - Special effects
  - Storyboarding

Unit – V

- **Main**
  - Compression
  - Multimedia Architecture
  - Databases
- **Detail**
  - Lossless compression
  - Lossy compression
  - Image compression
  - Audio compression
  - Video compression
  - MPEG standards
  - Multimedia architecture
  - Multimedia databases